EduPredict

Student Academic Performance Analysis and Prediction Using Machine Learning

A Project Report for the subject **Predictive Analytics**

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Dataset: Student Performance Factors (Kaggle)
Tools: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Streamlit

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1. Abstract

Student exam results depend on many academic, personal and environmental factors. This project analyses the **Student Performance Factors** dataset from Kaggle (6,607 student records, 19 input variables and the target variable *Exam_Score*) to understand which factors are associated with examination performance and to build a model that predicts a student's expected exam score.

The data was cleaned (three columns with missing values, no duplicates, one impossible score of 101 removed), explored with univariate, bivariate and multivariate analysis, and investigated through five objectives. Four regression models were trained using the same 80/20 train-test split and scikit-learn pipelines: Linear Regression, Decision Tree, Random Forest and K-Nearest Neighbors. On the test set, **Linear Regression** performed best (R^2 = 0.825, MAE = 0.42 marks, RMSE = 1.52 marks), followed by Random Forest (R^2 = 0.728), KNN (R^2 = 0.648) and Decision Tree (R^2 = 0.602). The same ranking was found with 5-fold cross-validation.

Attendance (r = 0.58) and weekly study hours (r = 0.45) showed the strongest associations with exam score, while sleep duration showed none. The best model is deployed in a Streamlit application, **EduPredict**, that predicts the exam score from user inputs. All findings are associations and not proof of cause and effect.

Keywords: predictive analytics, regression, exam score prediction, exploratory data analysis, Linear Regression, Random Forest, Streamlit.

2. Introduction

Predictive analytics uses historical data to estimate future outcomes. In education, it can help identify students who may need support before the final examination. Many factors are believed to affect results: how much a student studies, how regularly they attend, their earlier marks, their sleep, access to tutoring, family support and motivation. This project uses real recorded data to test which of these beliefs are supported and to build a regression model for predicting exam scores.

3. Problem Statement

To analyse the factors associated with students' examination performance and to develop a machine learning regression model that predicts a student's expected final examination score from information available before the examination, such as study hours, attendance, previous scores, sleep hours, tutoring sessions, motivation and family and school conditions.

4. Project Objectives

- **Objective 1:** Study the relationship between hours studied and exam score.
- **Objective 2:** Analyse how attendance is associated with exam score.
- **Objective 3:** Investigate the relationship between previous scores and exam score.
- **Objective 4:** Study how sleep hours and tutoring sessions are associated with exam score.
- **Objective 5:** Investigate how motivation level, parental involvement and access to resources are related to exam score.
- **Objective 6 (modelling):** Build and compare four regression models, select the best-fit model on test-set performance, and deploy it in a prediction interface.

5. Dataset Description

The dataset **Student Performance Factors** (file *StudentPerformanceFactors.csv*) was obtained from Kaggle (https://www.kaggle.com/datasets/lainguyn123/student-performance-factors). It has **6,607 rows and 20 columns**: 6 numerical inputs, 13 categorical inputs and the numerical target *Exam_Score*. The real file was used; no data was generated or invented.

Column	Type	Meaning
Hours_Studied	Numeric	Study hours per week
Attendance	Numeric	Class attendance percentage (60-100)
Parental_Involvement	Categorical	Low / Medium / High
Access_to_Resources	Categorical	Availability of learning resources: Low / Medium / High
Extracurricular_Activities	Categorical	Yes / No
Sleep_Hours	Numeric	Average sleep hours per night (4-10)
Previous_Scores	Numeric	Score in previous exams (50-100)
Motivation_Level	Categorical	Low / Medium / High
Internet_Access	Categorical	Yes / No
Tutoring_Sessions	Numeric	Tutoring sessions per month (0-8)
Family_Income	Categorical	Low / Medium / High
Teacher_Quality	Categorical	Low / Medium / High (78 missing)
School_Type	Categorical	Public / Private
Peer_Influence	Categorical	Positive / Neutral / Negative
Physical_Activity	Numeric	Hours of physical activity per week (0-6)
Learning_Disabilities	Categorical	Yes / No
Parental_Education_Level	Categorical	High School / College / Postgraduate (90 missing)
Distance_from_Home	Categorical	Near / Moderate / Far (67 missing)
Gender	Categorical	Male / Female
Exam_Score	Numeric (target)	Final exam score (55-101 in raw data)

6. Tools and Technologies

Table 2: *Tools and technologies*

Tool	Use in this project
Python 3	Programming language
Pandas, NumPy	Loading, cleaning and analysing data
Matplotlib, Seaborn	All graphs
Scikit-learn	Pipelines, imputation, encoding, scaling, four regression models, metrics, cross-validation
SciPy	Correlation and ANOVA p-values (installed with scikit-learn)
Joblib	Saving and loading the trained model
Streamlit	EduPredict prediction interface
Jupyter / Google Colab	Running the analysis notebook

7. Methodology

The project followed eight phases: (1) dataset download and setup; (2) understanding and preprocessing; (3) exploratory data analysis and the five objectives; (4) feature engineering; (5) building four models; (6) evaluation and best-model selection; (7) the prediction interface; (8) reporting. The data was split once (80 percent training, 20 percent testing, random_state = 42) and the same split was used for all models. All preprocessing for modelling was placed in scikit-learn pipelines so it was fitted on the training data only, preventing data leakage.

8. Data Preprocessing

8.1 Missing values

Table 3: *Missing values and how they were handled*

Column	Missing	Percent	Strategy
Teacher_Quality	78	1.18	Most frequent value (Medium)
Parental_Education_Level	90	1.36	Most frequent value (High School)
Distance_from_Home	67	1.01	Most frequent value (Near)

The three columns are categorical and only about 1 percent each is missing, so the most frequent category was used. Dropping the 229 incomplete rows would waste data. For the models, imputation is part of the pipeline (SimpleImputer) and is learned from the training set only.

8.2 Duplicates and inconsistent values

No duplicate rows were found, so none were removed. All categorical columns were checked for spelling, case and spacing differences and were consistent. Numerical ranges were checked against valid limits (percentages 0-100, scores 0-100, sleep 0-24 hours). The only invalid value was **one record with Exam_Score = 101**, which is impossible on a 100-mark scale, so that single row was removed (6,606 rows remain).

8.3 Outliers

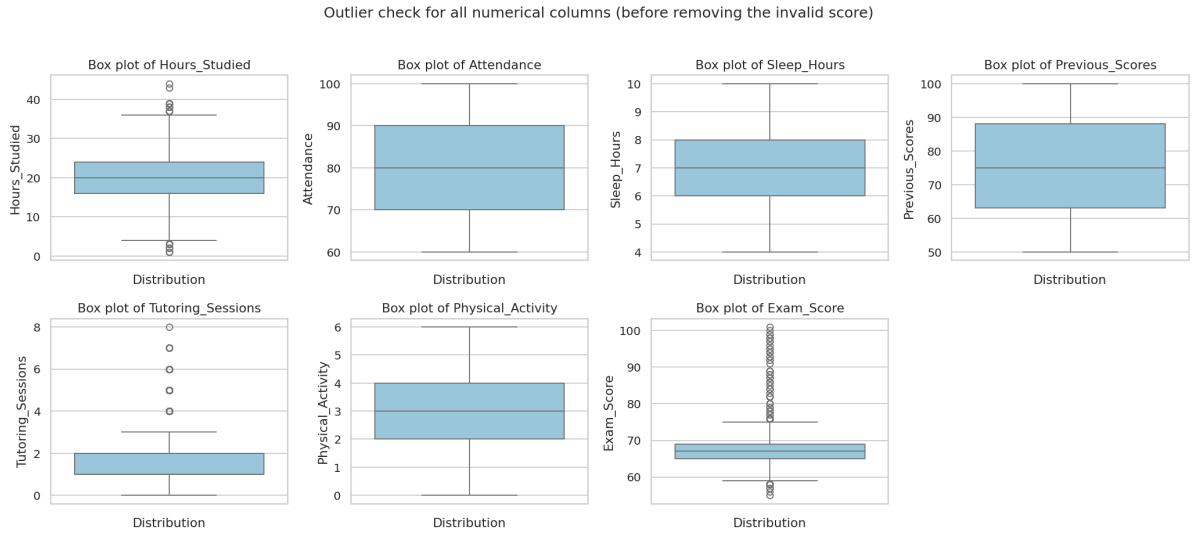


Figure 1: Box plots of all numerical columns (before removing the invalid score)

The IQR method (values beyond 1.5 times the IQR from the quartiles) flagged 43 values in Hours_Studied, 430 in Tutoring_Sessions and 104 in Exam_Score. These are realistic values (for example 4 or more tutoring sessions or a score of 90) and were **retained**; IQR shows that a value is unusual, not wrong.

8.4 Encoding, scaling and splitting

Categorical variables were encoded with OneHotEncoder (unknown categories ignored). Numerical variables were standardised with StandardScaler for Linear Regression and KNN (KNN is distance-based and needs all features on a similar scale); tree-based models were not scaled. The

data was split into 5,284 training and 1,322 test rows.

9. Exploratory Data Analysis

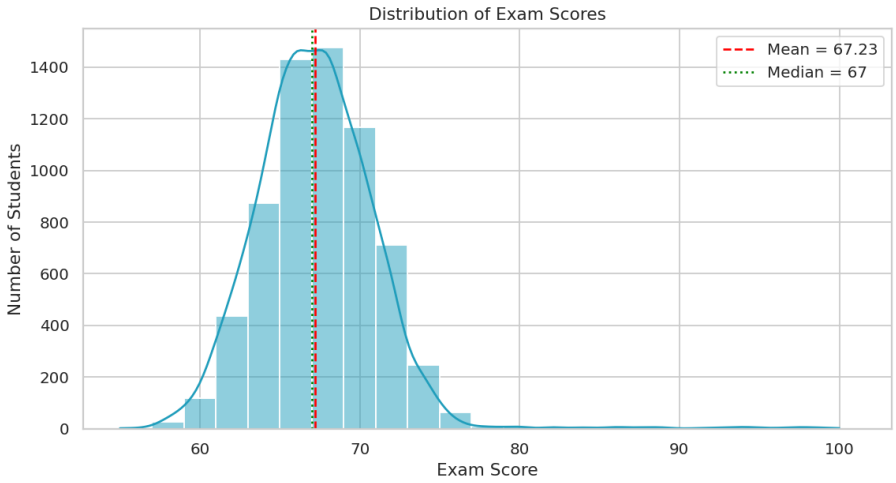


Figure 2: Distribution of exam scores

Exam scores are concentrated around the mean of 67.2 (standard deviation 3.9). Most students score between 65 and 69, with a right tail up to 100.

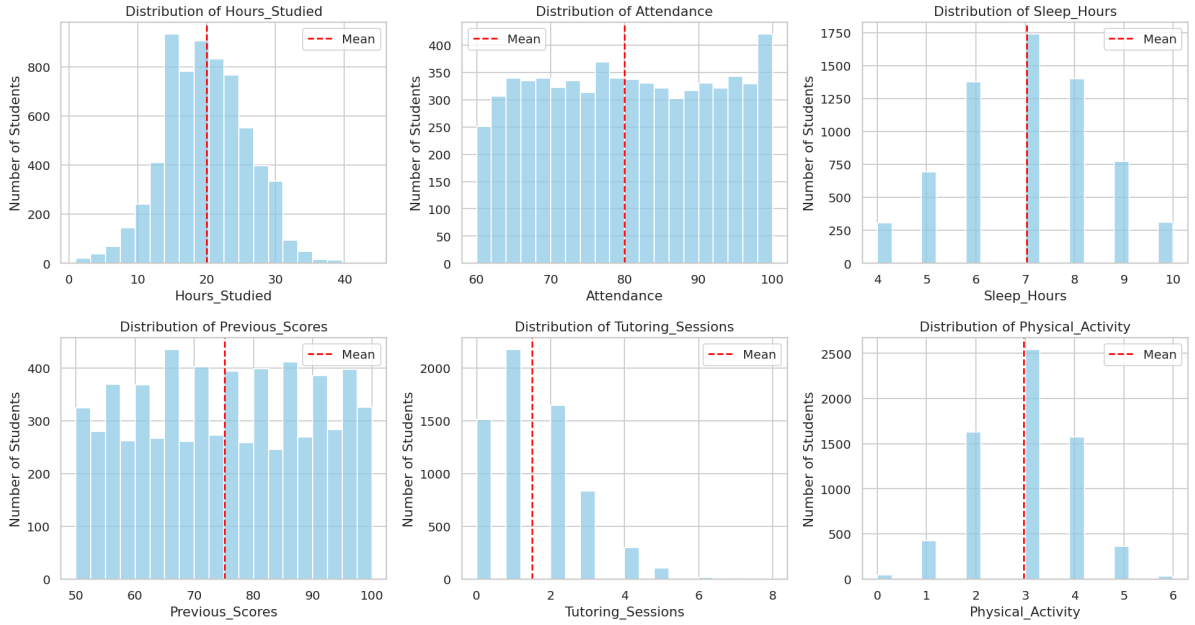


Figure 3: Distributions of the numerical features

Hours studied is bell-shaped around 20 hours, attendance and previous scores are spread across their ranges, sleep clusters near 7 hours, and most students have 0 to 2 tutoring sessions.

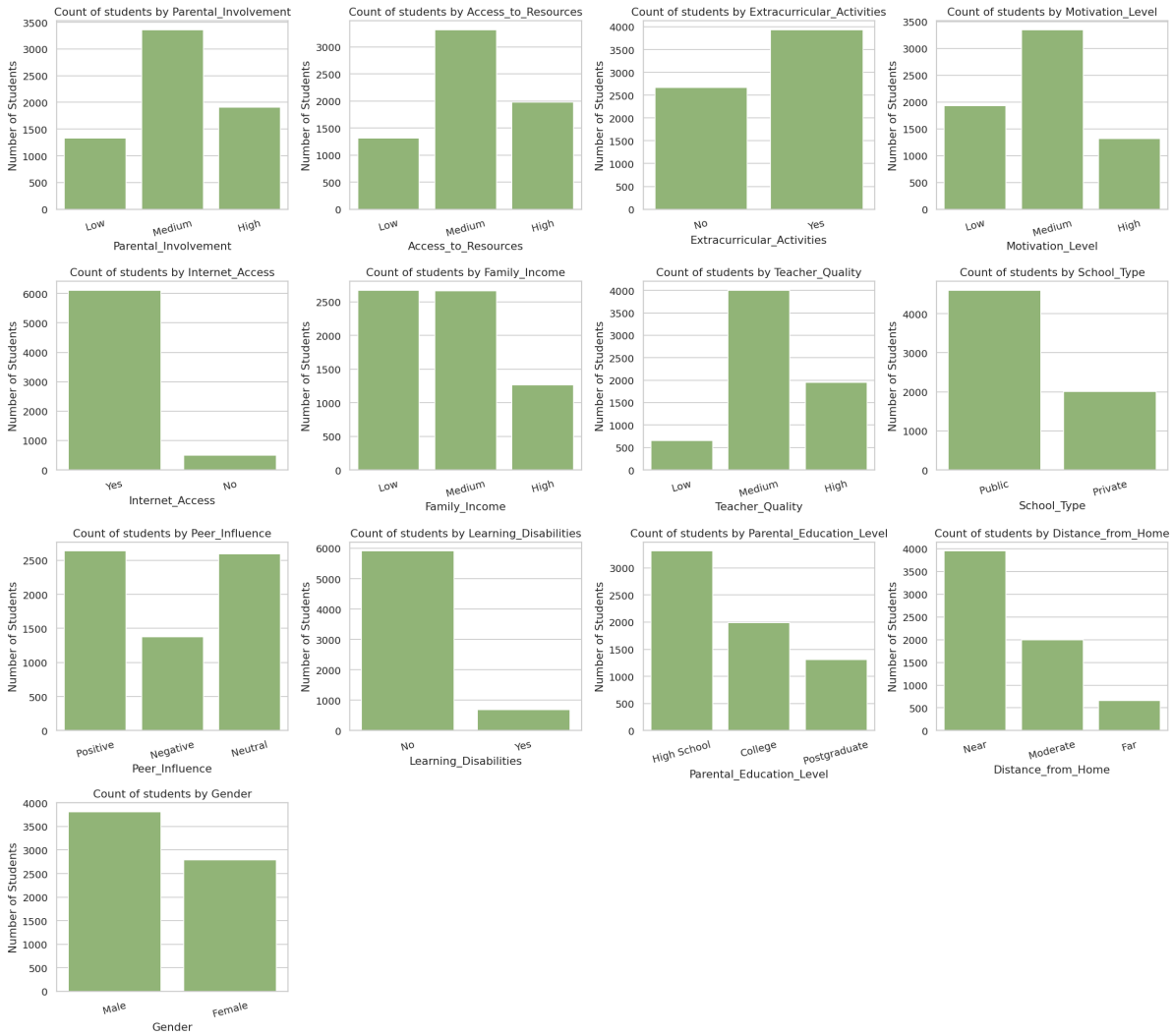


Figure 4: Count of students in each category of the categorical features

Most students have Medium parental involvement and resources; about 92 percent have Internet access, about 90 percent have no learning disability and about 70 percent attend public schools.

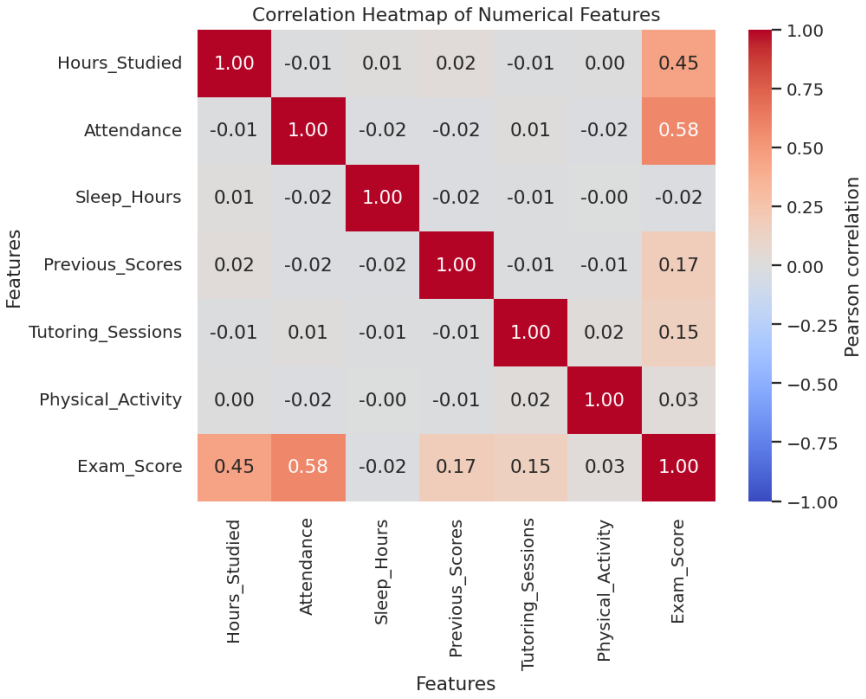


Figure 5: Correlation heatmap of the numerical features

Correlation with exam score: Attendance 0.58, Hours_Studied 0.45, Previous_Scores 0.17, Tutoring_Sessions 0.15, Physical_Activity 0.03, Sleep_Hours -0.02. The inputs are almost uncorrelated with each other. Correlation shows linear association only, not cause.

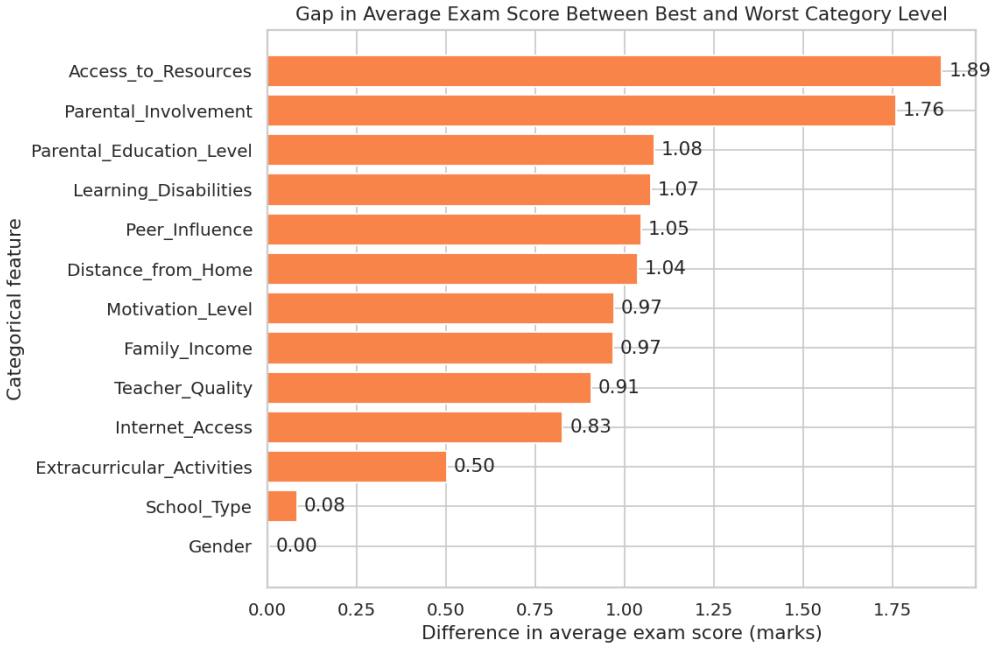


Figure 6: Difference in average exam score between the best and worst level of each categorical feature

Access to resources (1.89 marks) and parental involvement (1.76 marks) show the largest gaps. Gender (0.00) and school type (0.08) show practically none.



Figure 7: Pair plot of the main numerical features coloured by motivation level (1,000-student sample)

Exam score rises visibly with study hours and attendance and only weakly with previous scores. The predictors are unrelated to each other.

10. Objective-wise Analysis

Objective 1: Study hours

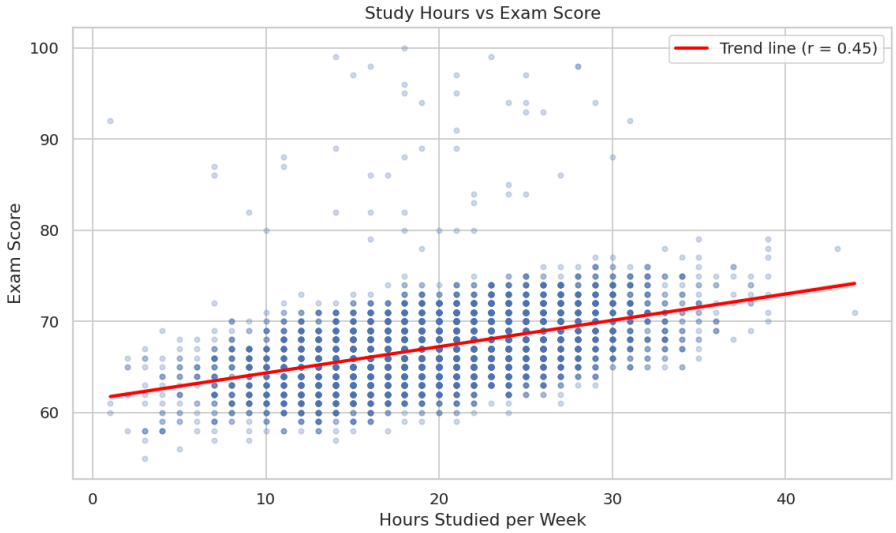


Figure 8: Study hours versus exam score with trend line

Result: Pearson r = 0.447 (p < 0.001); fitted line: Exam_Score = 61.47 + 0.288 x Hours_Studied. Each extra weekly hour is associated with about 0.29 more marks. **Conclusion:** students who study more tend to score higher; study hours account for about 20 percent of score variation on their own.

Objective 2: Attendance

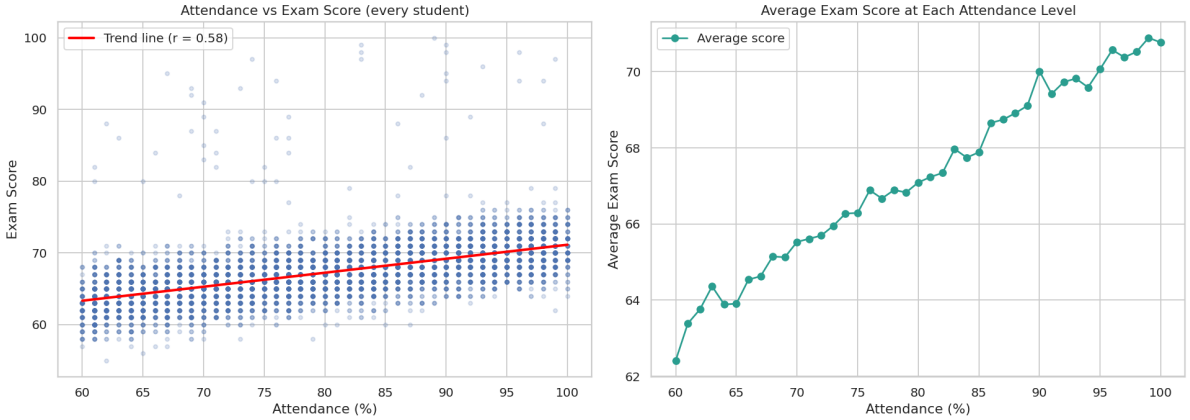


Figure 9: Attendance versus exam score (all students and average by attendance level)

Result: r = 0.582, the strongest of all features; fitted line: Exam_Score = 51.62 + 0.195 x Attendance. Average score is 62.4 at 60 percent attendance and 70.8 at 100 percent. **Conclusion:** higher attendance is strongly associated with higher scores.

Objective 3: Previous scores

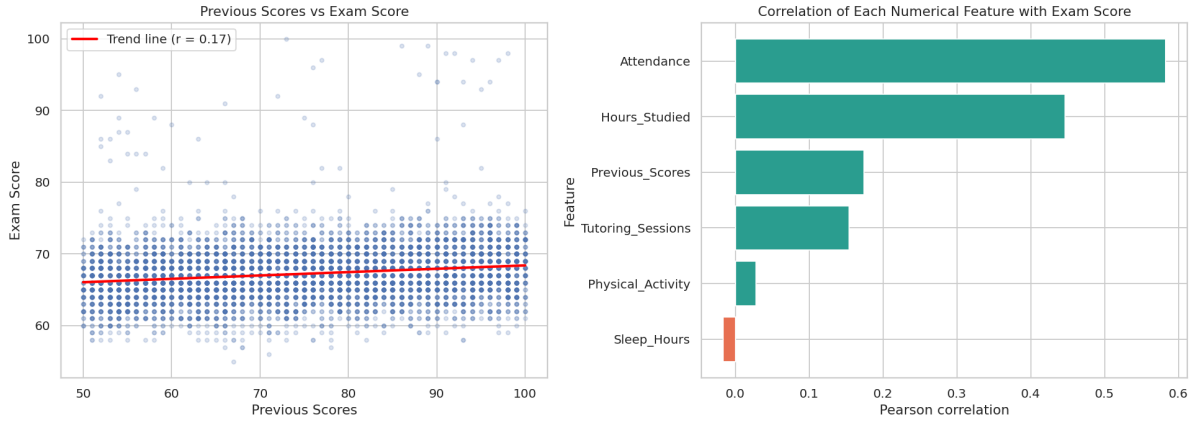


Figure 10: Previous scores versus exam score, and correlation of every numerical feature with exam score

Result: r = 0.174; each extra previous mark is associated with only 0.047 marks. **Conclusion:** in this dataset past scores are a weak predictor compared with attendance and study hours.

Objective 4: Sleep and tutoring

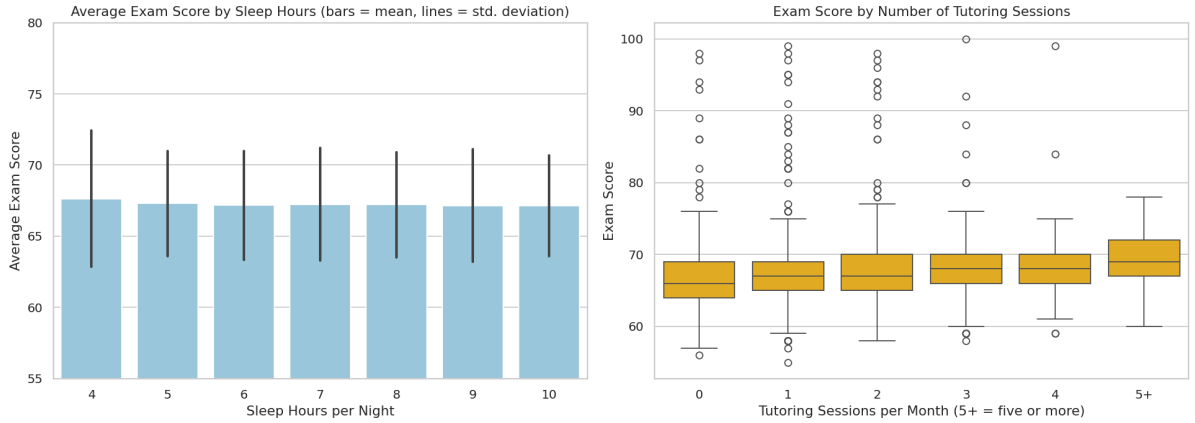


Figure 11: Average score by sleep hours, and score distribution by number of tutoring sessions

Table 4: *Average exam score by tutoring sessions*

Tutoring sessions	0	1	2	3	4	5+
Average exam score	66.49	66.98	67.57	67.89	68.23	69.23
Students	1,513	2,179	1,649	836	301	128

Result: Sleep: r = -0.016 (p = 0.19, not significant), averages are about 67 for every sleep duration. Tutoring: r = 0.154 (p < 0.001), with scores increasing from 66.5 to 69.2. **Conclusion:** sleep duration shows no association with score in this data; tutoring has a small positive association. Neither is proof of cause.

Objective 5: Motivation, parental involvement and resources

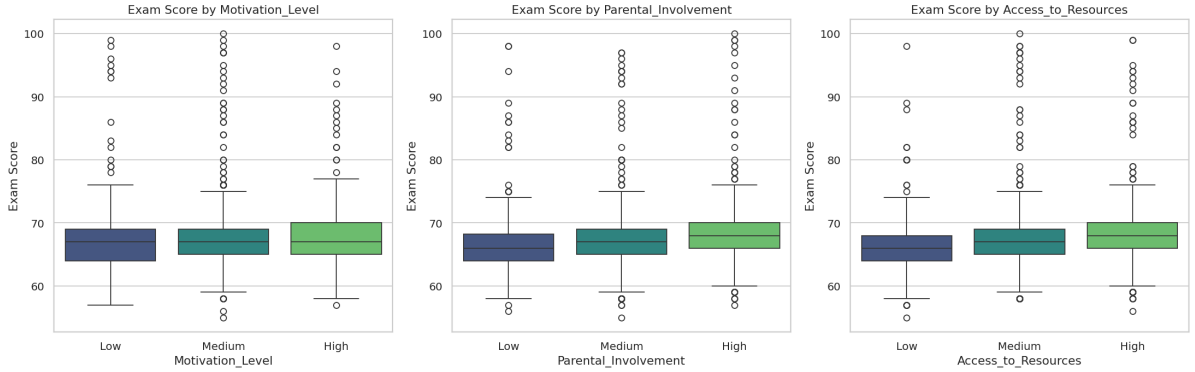


Figure 12: Exam score by motivation level, parental involvement and access to resources

Table 5: *Average exam score by level*

Factor	Low	Medium	High	High minus Low	ANOVA p
Motivation_Level	66.73	67.33	67.70	0.97	1.8e-12
Parental_Involvement	66.33	67.10	68.09	1.76	2.8e-38
Access_to_Resources	66.20	67.12	68.09	1.89	2.9e-43

Conclusion: students at the High level of each factor score higher on average, and the differences are statistically significant, but they are small (about 1 to 2 marks) and the distributions overlap heavily. The effect linked to attendance (about 8 marks across its range) is much larger.

11. Feature Engineering

Three category features were created for analysis: **Study_Category** (Low up to 15 h, Moderate 16-24 h, High 25+ h, based on the quartiles), **Attendance_Category** (Low 60-69, Medium 70-84, High 85-100) and **Sleep_Category** (Short 4-5 h, Adequate 6-8 h, Long 9-10 h). They were calculated with pandas *cut* from the original columns, make groups easier to read and compare, and use no information from the exam score. Because they only repeat information from the numerical columns, the **original numerical columns were retained for model training** and the new columns were not used as model inputs.

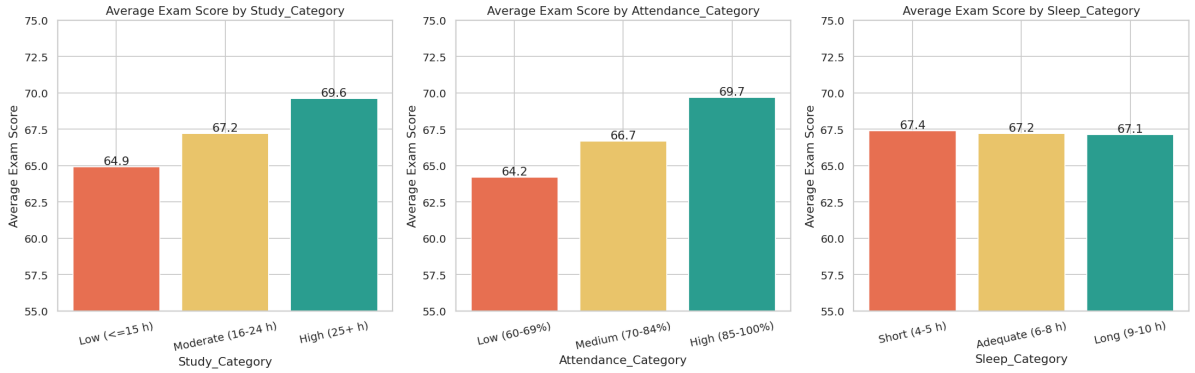


Figure 13: Average exam score by the engineered category features

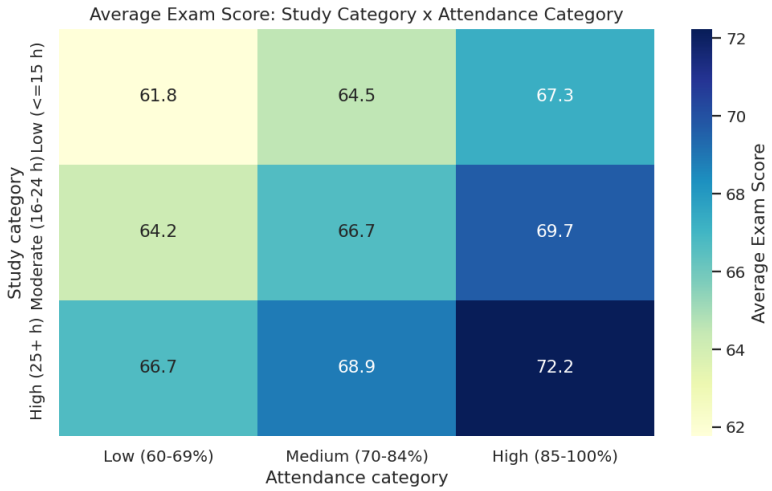


Figure 14: Average exam score for each combination of study and attendance category

The effects add up: students with high study and high attendance average 72.2 marks, versus 61.8 for low study and low attendance (about 10 marks apart). Sleep categories differ by less than 0.3 marks.

12. Machine Learning Algorithms

- **Linear Regression:** fits a weighted sum of the inputs (score = b0 + b1 x Hours + b2 x Attendance + ...) that minimises squared error. Simple, fast and easy to interpret; limited to straight-line relationships.
- **Decision Tree Regressor:** repeatedly splits students by yes/no questions on feature values; the prediction is the average score of the final group. Easy to visualise; can overfit if too deep.

- **Random Forest Regressor:** averages the predictions of many (200) decision trees built on random samples of rows and features, which reduces overfitting.
- **K-Nearest Neighbors Regressor:** predicts the average score of the K most similar students in the training data. Needs scaling; can be weak with many one-hot columns.

13. Model Training

Inputs X were the 19 original features and the target y was Exam_Score. The data was split 80/20 with random_state = 42 and the same split was used for all four models. Each model sits in a pipeline with a ColumnTransformer: numerical columns use median imputation (and standard scaling for Linear Regression and KNN); categorical columns use most-frequent imputation and one-hot encoding. Hyperparameters were chosen with 5-fold cross-validation on the training set only: Decision Tree (max_depth 15, min_samples_leaf 20, from depths 3-15), KNN (20 neighbours, from 5-50) while Random Forest used 200 trees with min_samples_leaf = 3 and Linear Regression has no tuning settings.

14. Model Evaluation

R² is the share of score variation explained by the model (higher is better). **MAE** is the average absolute error in marks. **MSE** is the average squared error (penalises large mistakes). **RMSE** is the square root of MSE, in marks. All were computed on the 1,322 test students.

Table 6: *Test-set performance of the four models*

Model	R²	MAE	MSE	RMSE	Train R²
Linear Regression	0.8250	0.4160	2.3142	1.5213	0.7110
Decision Tree	0.6023	1.4468	5.2596	2.2934	0.6486
Random Forest	0.7283	1.0234	3.5928	1.8955	0.8320
KNN	0.6482	1.3469	4.6518	2.1568	0.6014

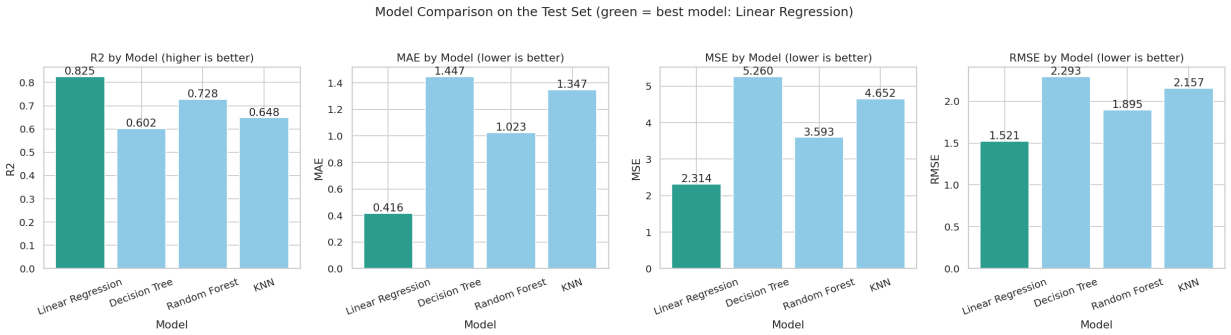


Figure 15: Model comparison on the four metrics (green = best model)

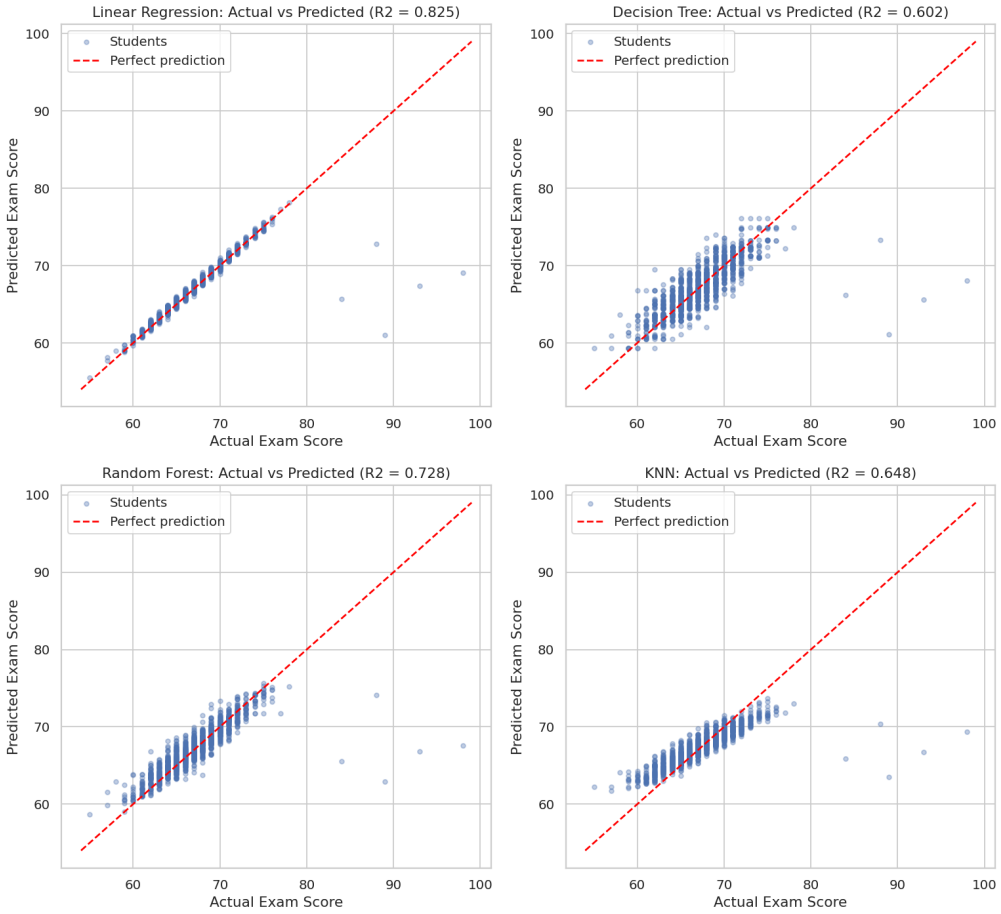


Figure 16: Actual versus predicted exam scores for the four models

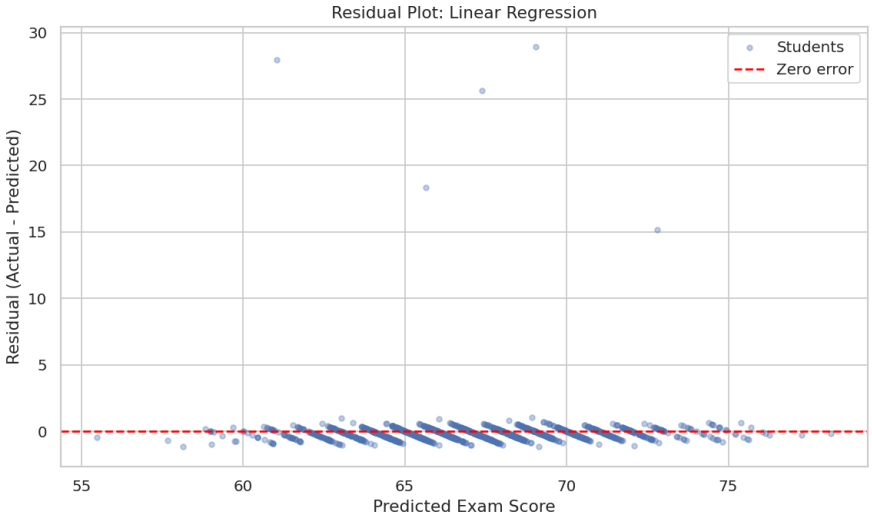


Figure 17: Residual plot for the best model (Linear Regression)

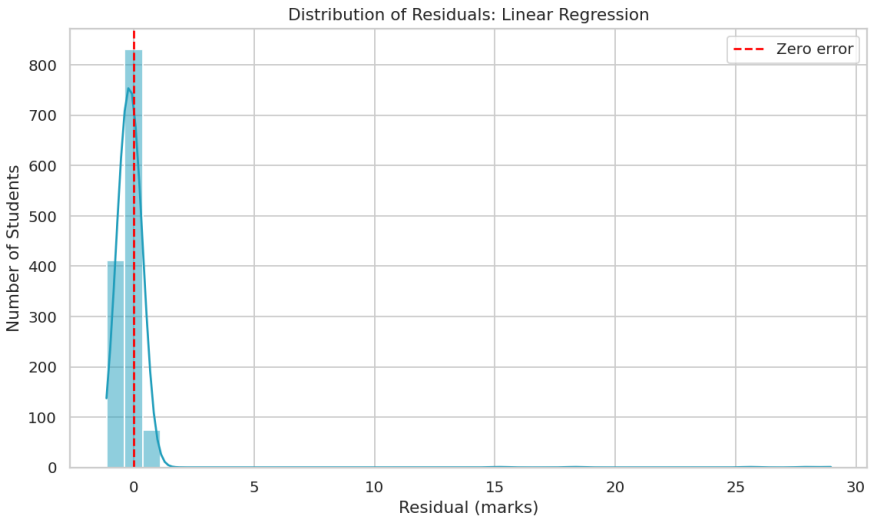


Figure 18: Distribution of residuals for the best model

For Linear Regression 98.7 percent of test predictions are within 1 mark. The residuals are centred near zero with a few large positive errors (students who scored far higher than predicted).

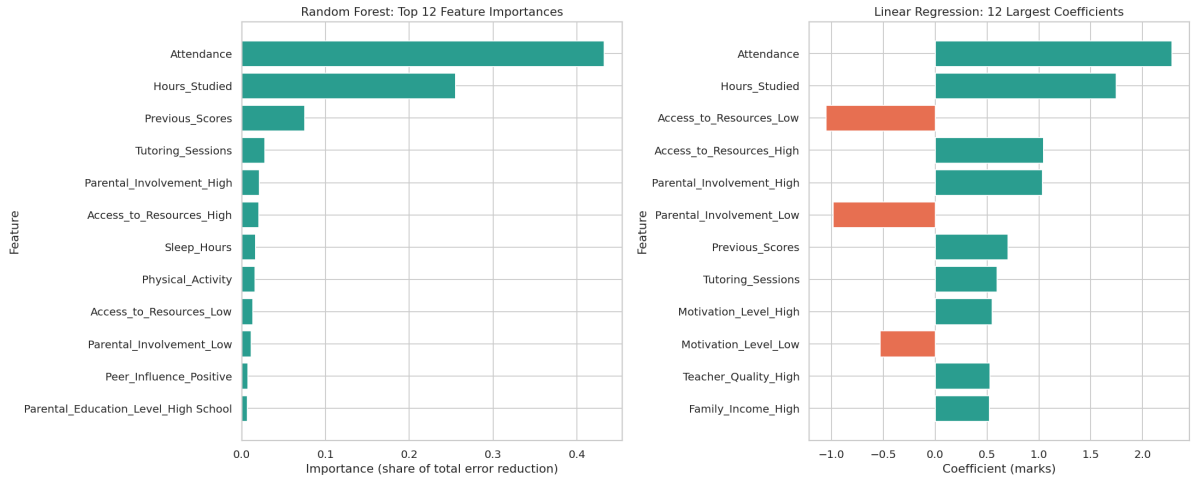


Figure 19: Random Forest feature importance and Linear Regression coefficients

Both models rank Attendance first and Hours_Studied second (Random Forest importance 43 percent and 26 percent). Next come previous scores, tutoring, parental involvement, access to resources and motivation. Sleep and physical activity carry almost no importance.



Figure 20: Summary dashboard: accuracy, error, closeness of predictions, key factors and main relationships

The dashboard summarises the project on one page: model accuracy (R^2 as a percentage), error in marks, the share of predictions within 1, 2 and 5 marks of the real score (98.7 percent within 1 mark for Linear Regression), the best model's actual-versus-predicted and residual plots, and the main factors.

15. Model Comparison

Table 7: *Test R^2 and cross-validated R^2 (training data)*

Model	Test R^2	5-fold CV R^2 (mean)	CV std
Linear Regression	0.8250	0.7166	0.0910
Decision Tree	0.6023	0.5314	0.0679
Random Forest	0.7283	0.6342	0.0798
KNN	0.6482	0.5574	0.0645

Best-fit model: Linear Regression, selected only from the test-set results (highest R^2, lowest MAE, MSE and RMSE) and confirmed by cross-validation. The relationships in this dataset are mostly straight-line and additive, which is the structure Linear Regression is designed for. Decision Tree and KNN approximate a straight line with steps or neighbourhood averages and need much more data to match it; Random Forest improves on the single tree by averaging but remains piecewise. Random Forest shows mild overfitting (train R^2 0.83 versus test 0.73). This conclusion applies to this dataset; on more complex real data, tree-based models often perform better.

16. Results and Discussion

Error analysis. For 6,516 of the 6,606 students (98.6 percent) the best model predicts within 1 mark, and no student has an error between 2 and 5 marks. However, **49 students (0.74 percent)** scored between 78 and 100 while the model predicted 60 to 76, an error of over 10 marks; none of the recorded features explains their scores. These few cases limit R^2: with the 5 such students in the test set excluded, test R^2 rises from 0.825 to 0.985. Only 5 of them fell into the test set (versus 44 in the training set), which is why the test R^2 is higher than the cross-validated R^2 (0.717). A realistic expectation for R^2 on new samples is therefore about 0.7 to 0.8, with a typical error well under 1 mark for most students.

Main insights. Attendance and study hours are the most strongly associated with score; tutoring, parental involvement, access to resources and motivation add smaller amounts; sleep, gender and school type show no meaningful association; previous scores matter only weakly in this dataset.

17. Conclusion

The project cleaned and explored the Student Performance Factors dataset, investigated five objectives and compared four regression models. Linear Regression was the best-fit model (R^2 = 0.825, MAE = 0.42, RMSE = 1.52 on the test set). Attendance and study hours emerged as the most important factors. The trained pipeline is stored in *models/best_model.pkl* and used by the EduPredict Streamlit application, which accepts validated user input and returns the expected exam score from the real trained model.

18. Recommendations

For students

- Attend classes regularly; attendance had the strongest association with scores.
- Keep a steady weekly study routine.
- Use tutoring when available; it showed a small positive association.
- Do not treat sleep duration as a score lever in this data, but do not neglect sleep for health reasons (the dataset has no sleep-quality information).

For educational institutions

- Track attendance as an early-warning indicator and follow up with students whose attendance is low.
- Encourage parental involvement and ensure access to learning resources for all students.
- Offer tutoring support, especially to students with low attendance and low study time.
- Use predictions only to offer support, never to label or penalise students.

19. Limitations

- Findings are **associations, not causes**.
- The data is unusually regular (tidy ranges, nearly independent features, and a near-linear relation for 99 percent of students), which suggests it may be simulated or heavily processed; results may not transfer to real classrooms.
- About 0.7 percent of students have scores unexplained by any feature; they were kept because they could not be proven wrong.
- Single exam score; no subject, time, or data-collection details; sleep measured in hours only.
- Only light hyperparameter tuning was done, to keep the project simple.

20. Future Scope

- Use real institutional data with more features such as assignment marks, subject and semester.
- Try tuned gradient-boosting models and compare them using cross-validation.
- Provide prediction ranges, not only a single value.
- Check fairness across student groups.
- Deploy the Streamlit app online for students and teachers.

21. Rubric Compliance

Table 8: *Teacher rubric compliance*

#	Requirement	How it was completed
1	Relevant dataset from Kaggle	Student Performance Factors, 6,607 rows, 20 columns (Section 5)
2	Preprocessing and cleaning	Missing values imputed, 0 duplicates, categories checked, invalid score removed, outliers reviewed (Section 8)
3	EDA with insights	Univariate, bivariate, multivariate analysis (Section 9)
4	At least five objectives	Five objectives plus modelling objective (Section 10)
5	Appropriate visualisation	20 labelled graphs including a one-page dashboard (Sections 8-14)
6	Feature engineering	Study, Attendance and Sleep categories (Section 11)
7	At least four ML models	Linear Regression, Decision Tree, Random Forest, KNN (Sections 12-13)
8	Best-fit model	Linear Regression, selected on test-set results and confirmed by cross-validation (Section 15)
9	Performance metrics	R^2, MAE, MSE, RMSE, residual analysis (Section 14)
10	Findings, conclusions, recommendations	Sections 16-18

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